

Background

- In Western populations epidermal growth factor receptor (EGFR) mutations occur in approximately 10% -15% of patients with non-small cell lung cancer (NSCLC).
- EGFR targeted tyrosine kinase inhibitors (TKIs) for NSCLC have revolutionized the treatment paradigm.
- Specialty clinical pharmacists play a crucial role in patient adherence to EGFR targeted TKI therapy.
- Patients with NSCLC at the integrated health system specialty pharmacy (HSSP) were enrolled in the lung cancer disease program with specific clinical services and disease management support tools.

Objective

- To describe the adherence, mutation patterns and treatment discontinuation rates among patients with NSCLC on EGFR targeted TKIs.

Methods



This is a retrospective chart review of completed activities between January 1st, 2024, to December 31st, 2024, for patients on EGFR TKIs as primary or subsequent treatment.



The primary outcome included evaluating the adherence (MPR and PDC) of TKIs for patients filling with the integrated HSSP.



Treatment discontinuation rates were evaluated for secondary outcomes.



Demographic data, presence of mutations, prescribing patterns of first-, second- and third generation TKIs were also evaluated.

Results

Table 1: Demographic data (n=96)

Sex at birth	
Male	27%
Female	73%
Average age (years)	
	74.5
Stages	
Stage I	8%
Stage II	4%
Stage III	9%
Stage IV	78%
Smoking Status	
Former smokers	48%
Never smokers	50%
Male	25%
Female	75%
Current	2%

Table 2: EGFR TKI Utilization for NSCLC

Osimertinib	83%
Afatinib	11%
Erlotinib	4%
Lazertinib	1%
Gefitinib	0%

Table 3: Mutations in NSCLC

Common*	81%
Uncommon**	15%
Combination	4%

*Common mutations include exon 19 deletions or exon 21 L858R substitution

**Uncommon mutations include S768I, L861Q, G719X, MLH1, NRG1-CD74 fusion, ATR

Figure 1: Average MPR and PDC in EGFR TKI treated patients with NSCLC (n=96)

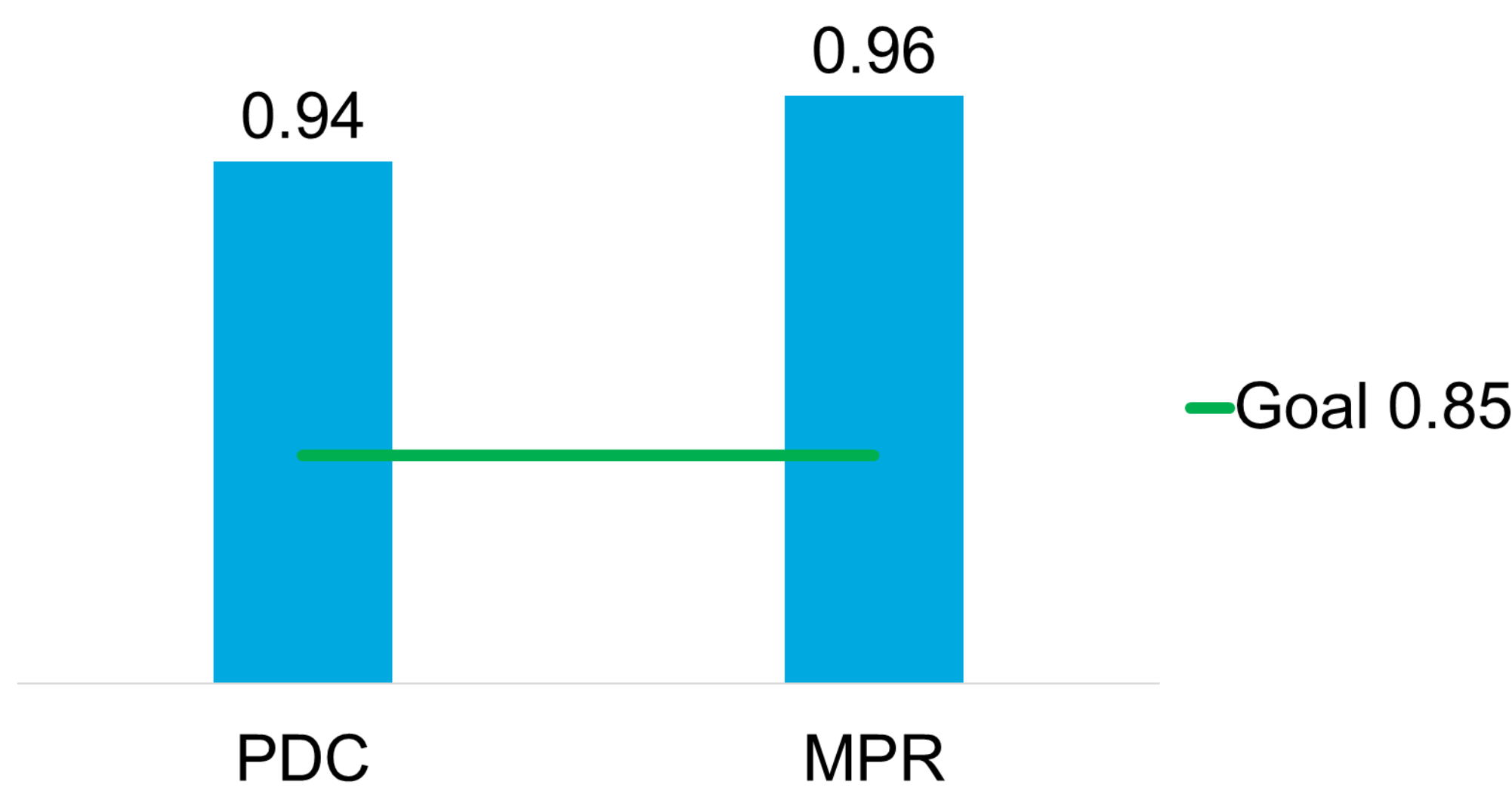
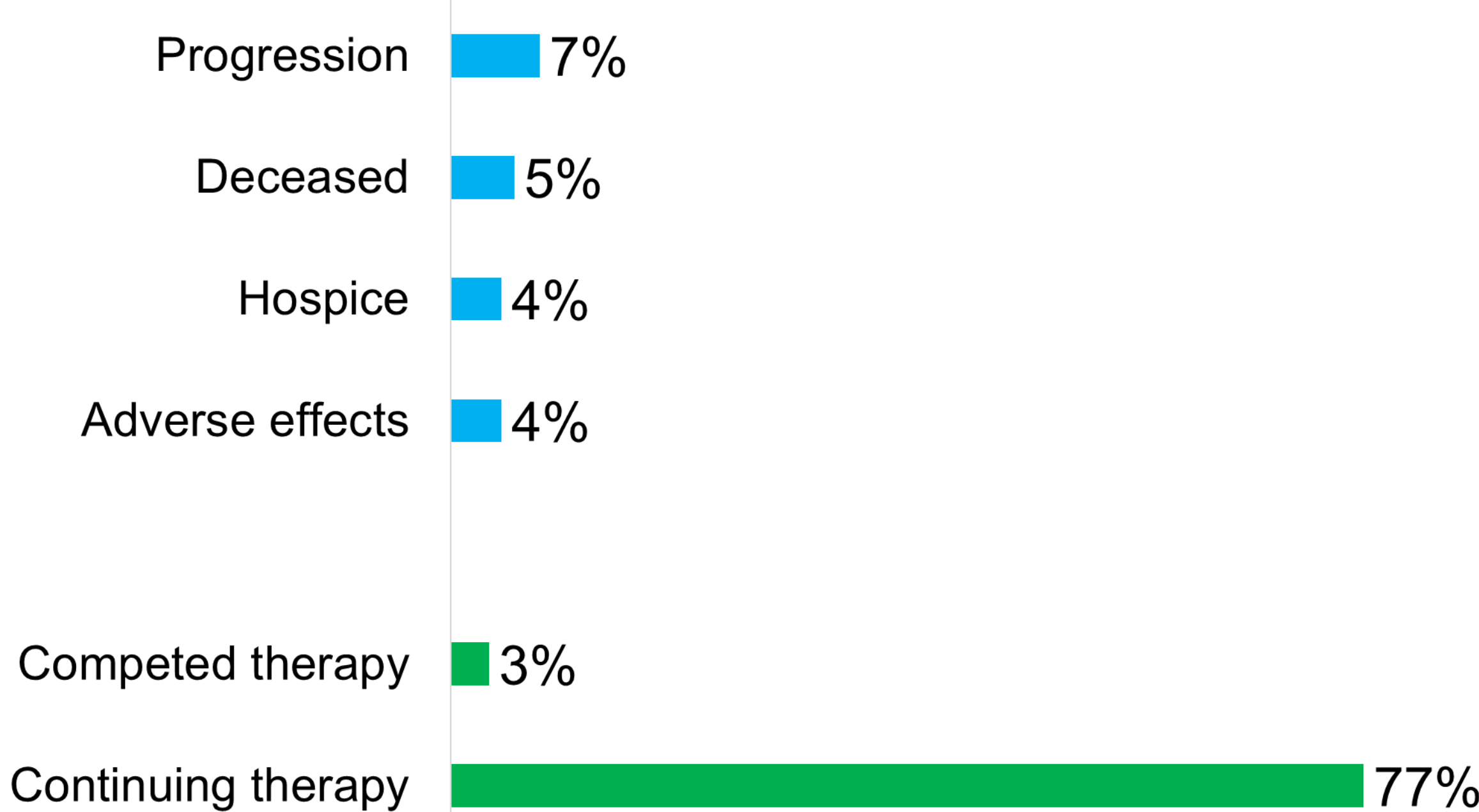
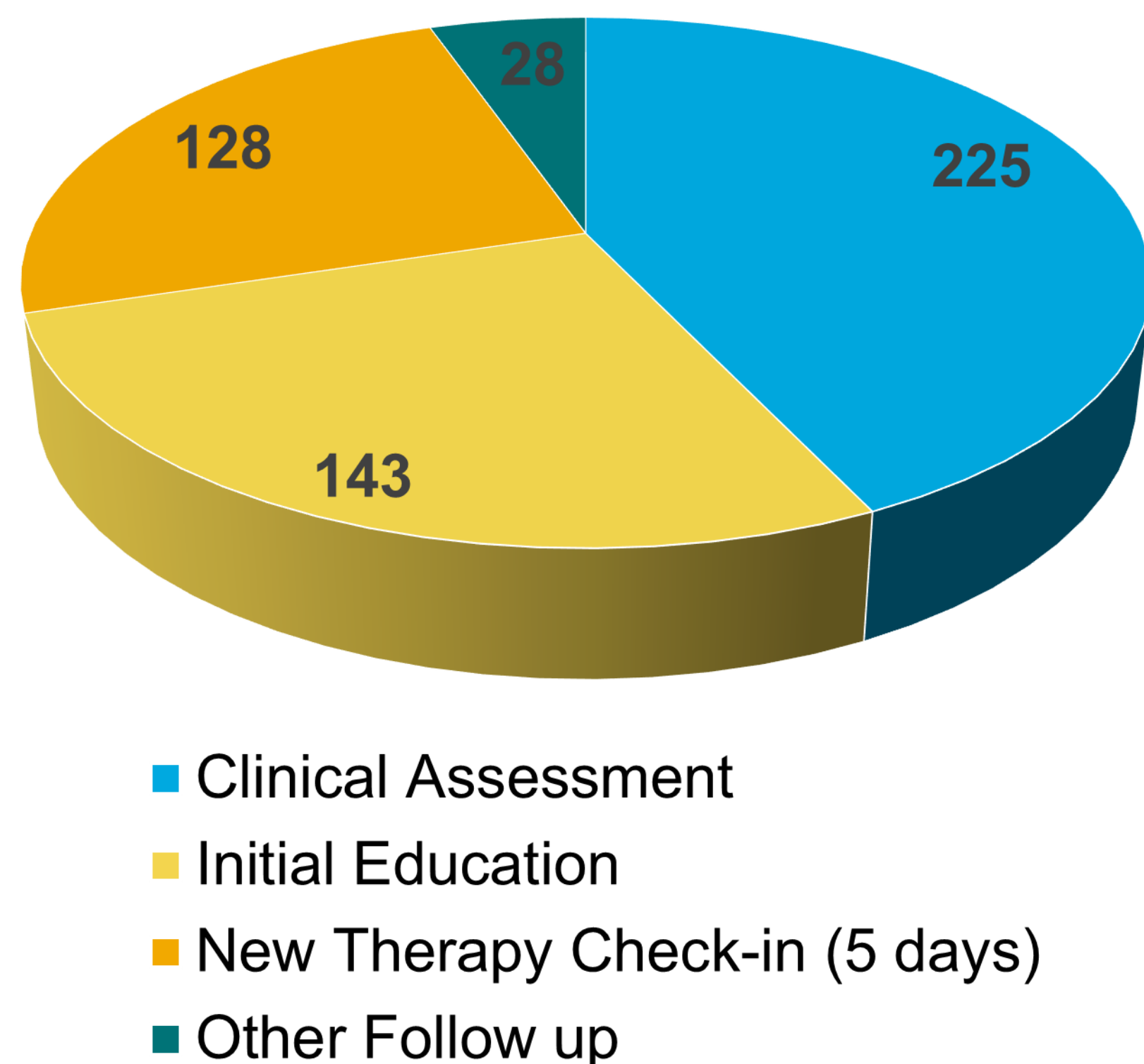


Figure 2: Treatment discontinuation rates (n=94)*



*Two patients who transitioned care during 2024 were excluded from this analysis

Figure 3: Pharmacist clinical services



Discussion

- The average age in this analysis was 74.5 years, consistent with previously reported literature.¹
- Among never-smokers, women are disproportionately affected- a trend observed in our cohort.²
- In patients with NSCLC, varying TKI discontinuation rates (7.7% to 18%) due to adverse events have been reported.^{3,4}
- In our analysis, 4% of patients discontinued EGFR TKIs due to adverse effects which is lower than the FLAURA trial. In the FLAURA trial, 13% of patients on osimertinib, and 18% of patients on standard EGFR TKIs discontinued therapy due to adverse events.³
- In 2024, on average specialty clinical pharmacists at the integrated HSSP had about 5.5 direct phone interactions per patient for assessments or initial education.
- The integrated HSSP model with enhanced patient engagement through frequent touchpoints along with therapy specific education increased medication adherence above benchmark as evidenced by MPR/PDC of 0.96/0.94.

Limitations

- Clinical trials enroll patients with well-defined characteristics. The methodology for reporting treatment discontinuation rates may vary among studies.
- Discontinuation rates observed in this analysis may not be directly comparable to clinical trials.

Conclusion

Integrated HSSPs are ideally positioned to track patient outcomes in NSCLC.

References

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The authors of this presentation have nothing to disclose concerning possible financial or personal relationships with commercial entities that may have a direct or indirect interest in the subject matter of this presentation.